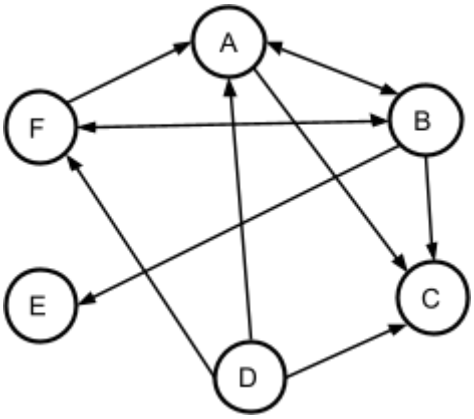
American Computer Science League

2023-2024 ● Contest 4: Short Problems ● Intermediate Division

1. Graph Theory

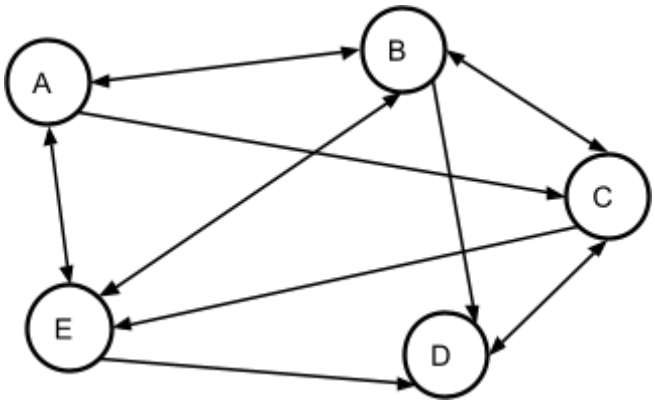
How many paths of length 2 are there in the following directed graph?



A. 12
B. 14
C. 15
D. 16
E. 18

2. Graph Theory

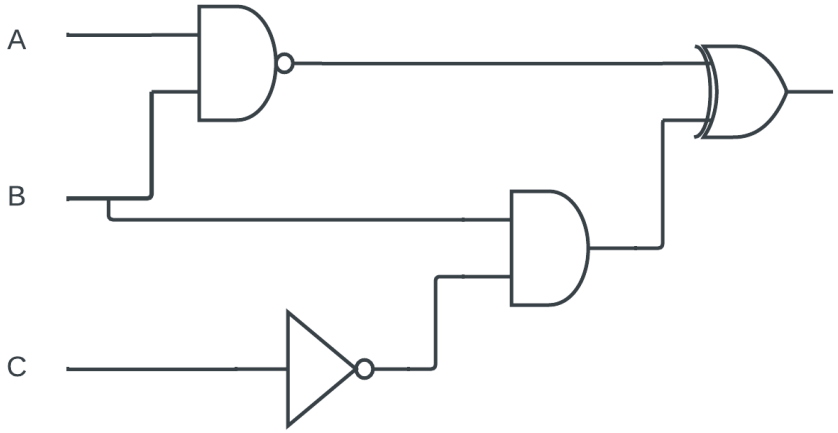
How many cycles are there in the following directed graph from vertex A?



A. 7
B. 8
C. 9
D. 11
E. 13

3. Digital Electronics

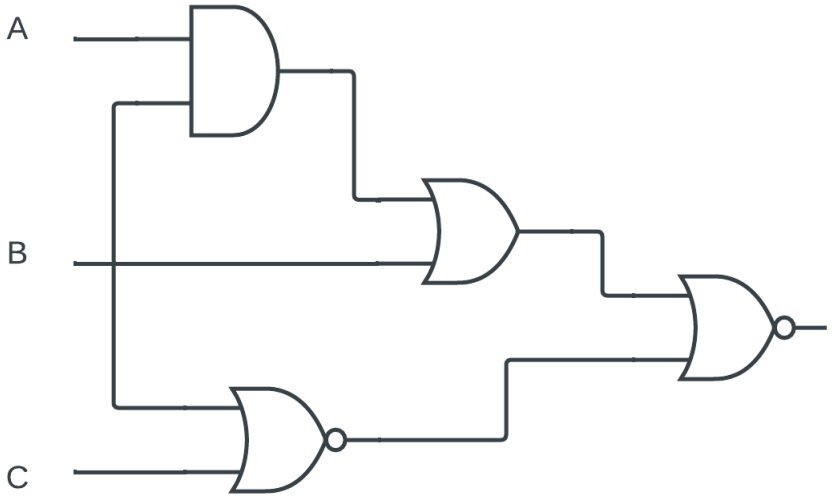
How many ordered triples make the following digital circuit TRUE?



A. 2
B. 4
C. 5
D. 6
E. 8

4. Digital Electronics

Simplify the Boolean expression represented by the digital diagram below:



A. $\overline{B}C$
B. $\overline{BC}$
C. $\overline{B} * \overline{C}$
D. $B\overline{C}$
E. BC

5. Assembly Language

What is the output when the following program is executed?

```
N       DC      1
T       DC      1
M       DC      78
        LOAD    T
TOP     SUB     M
        BE      STOP
        LOAD    N
        ADD     =1
        STORE   N
        LOAD    T
        ADD     N
        STORE   T
        BU      TOP
STOP    PRINT   N
        END
```

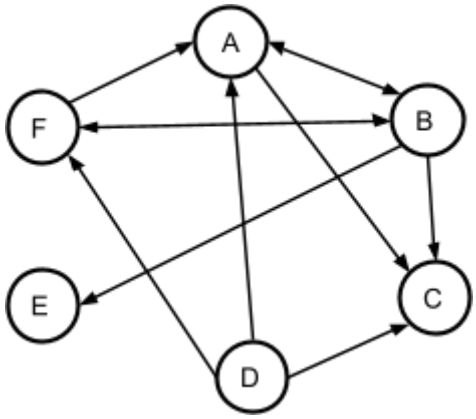
A. 11
B. 12
C. 13
D. 14
E. 15

American Computer Science League

2023-2024 ● Contest 4: Short Problems ● Intermediate Division

1. 图论

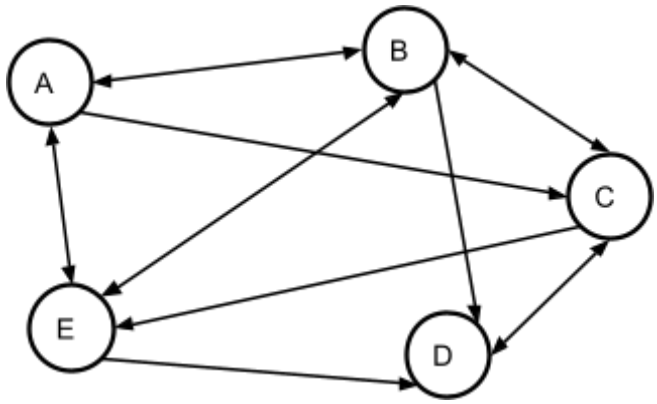
请问**在下列有向图中，有多少条长度为** 2 **的路径？**



A. 12
B. 14
C. 15
D. 16
E. 18

2. 图论

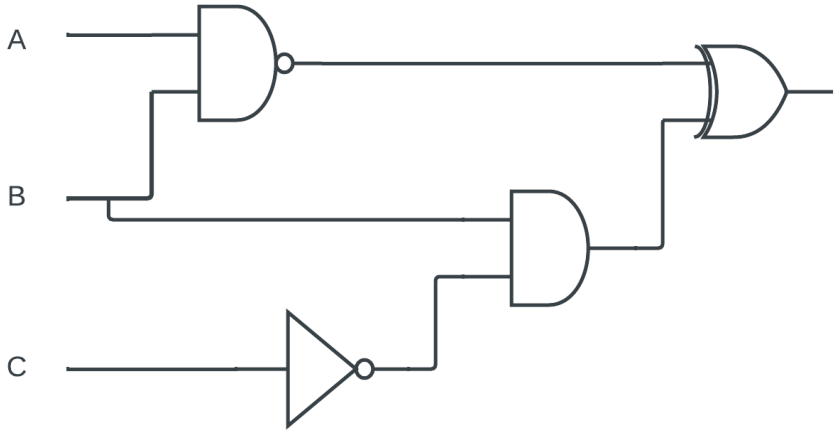
请问**在**下列有向图中，**从顶点** A 开始**有多少个环？**



A. 7
B. 8
C. 9
D. 11
E. 13

3. 数字电路

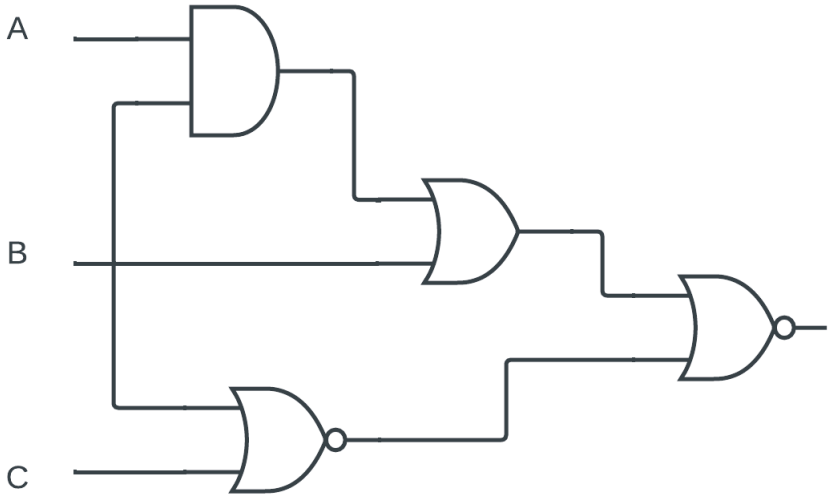
请问**有多少个有序三元组可以使得下述电路图为真？**



A. 2
B. 4
C. 5
D. 6
E. 8

4. 数字电路

简化**下述用数字电路图表达的布尔表达式：**



A. $\overline{B}C$
B. $\overline{BC}$
C. $\overline{B} * \overline{C}$
D. $B\overline{C}$
E. BC

5. 汇编语言

执行**下列**程序**时会输出什么？**

```
N         DC        1
T         DC        1
M         DC        78
          LOAD      T
TOP       SUB       M
          BE        STOP
          LOAD      N
          ADD       =1
          STORE     N
          LOAD      T
          ADD       N
          STORE     T
          BU        TOP
STOP      PRINT     N
          END
```

A. 11
B. 12
C. 13
D. 14
E. 15